

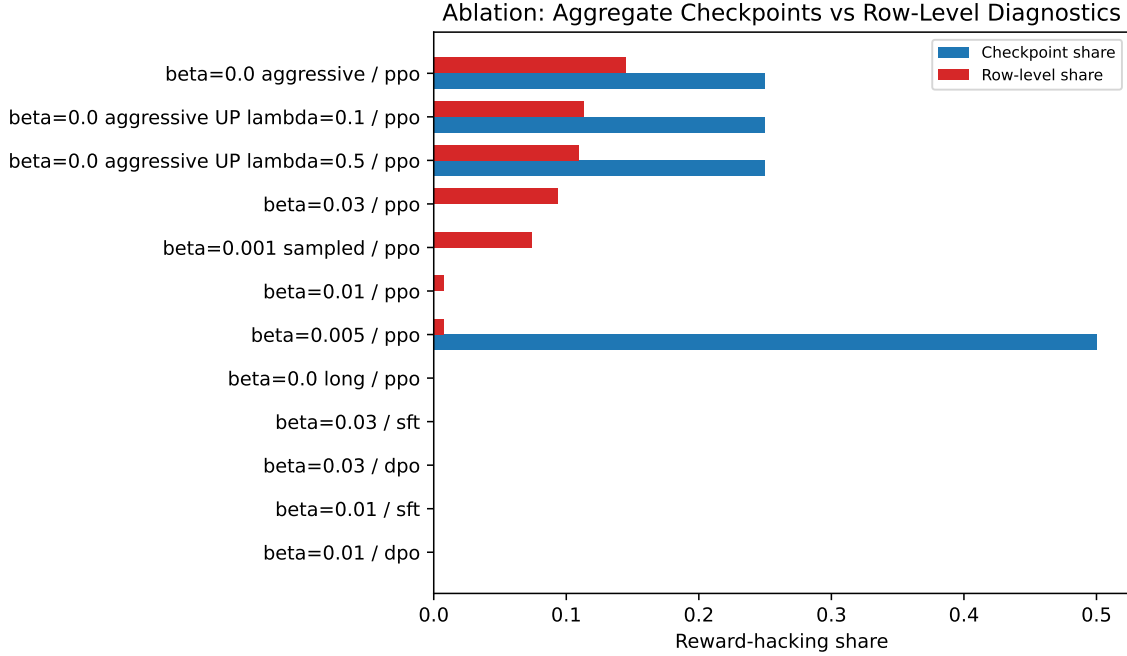


Figure 4: Ablation comparing checkpoint-level and row-level reward-hacking shares. Several settings contain row-level reward hacking even when checkpoint-level transitions do not classify as reward hacking.

directions. These examples are not used to define the taxonomy; they illustrate the local behaviors that the taxonomy exposes.

5.9 Judge disagreement is a separate diagnostic

Evaluator gaming appears in 14 of 31 checkpoint transitions (45.2%; bootstrap 95% CI: 29.0–61.3%) but only 75 of 1,920 row-level transitions (3.9%; bootstrap 95% CI: 3.1–4.8%). This difference suggests that aggregate judge disagreement can be amplified by small mean shifts even when most individual prompts do not show opposite judge movement. At row level, evaluator-gaming share is highest for proxy under-alignment (17.2%), stable alignment (15.0%), and reward hacking (11.8%). We did not perform repeated judge sampling or length-controlled judging; however, the stored length diagnostics do not support a simple verbosity-only explanation. Reward-hacking rows are longer on average than other rows, but both groups shorten across transitions, and the correlation between target length and reward-hacking status is small ($r = 0.083$). The result motivates treating judge disagreement as a distinct monitoring channel rather than folding it silently into $\bar{R}^\dagger$.

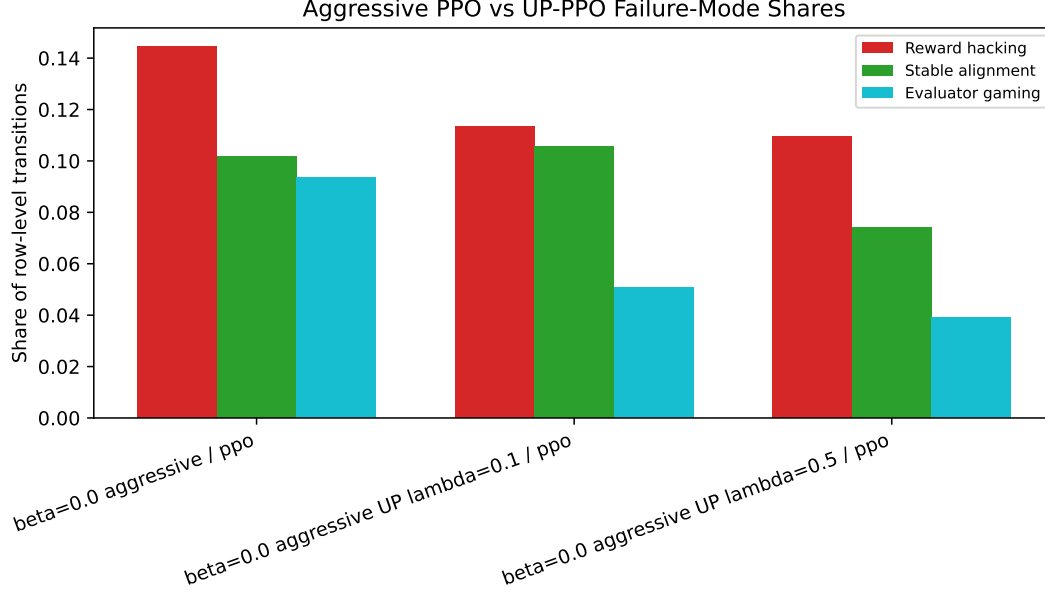


Figure 5: Aggressive PPO versus UP-PPO. UP-PPO reduces row-level reward-hacking and evaluator-gaming shares relative to the aggressive PPO condition, though localized failures remain.

Table 6: Top early-warning feature importances. Logistic-regression values are standardized coefficients; random-forest values are impurity importances. Transition-diagnostic rows are included only as a sanity check because these features overlap with the label definition.

Feature set	Model	Feature	Importance	Importance
Pre-state	Logistic	previous $R^\dagger$	0.959	0.959
Pre-state	Logistic	previous length	0.484	0.484
Pre-state	Logistic	previous distinct-1	0.447	0.447
Pre-state	Logistic	previous R_ϕ	-0.432	0.432
Pre-state	Logistic	previous 3-gram repetition	-0.339	0.339
Pre-state	Logistic	previous distinct-2	-0.285	0.285
Pre-state	Logistic	previous KL	-0.276	0.276
Pre-state	Logistic	previous judge disagreement	-0.210	0.210
Pre-state	Logistic	previous uncertainty	-0.163	0.163
Pre-state	Logistic	previous $R_2^\dagger$	-0.158	0.158
Pre-state	Random forest	previous $R^\dagger$	0.171	0.171
Pre-state	Random forest	previous KL	0.148	0.148
Pre-state	Random forest	previous R_ϕ	0.144	0.144
Pre-state	Random forest	previous length	0.142	0.142
Pre-state	Random forest	previous distinct-1	0.107	0.107
Pre-state	Random forest	previous distinct-2	0.100	0.100
Pre-state	Random forest	previous uncertainty	0.098	0.098
Transition	Logistic	$\Delta R^\dagger$	-3.613	3.613
Transition	Logistic	ΔR_ϕ	1.761	1.761
Transition	Logistic	$\Delta R_2^\dagger$	-0.376	0.376
Transition	Logistic	target distinct-2	0.173	0.173
Transition	Logistic	target 3-gram repetition	0.144	0.144

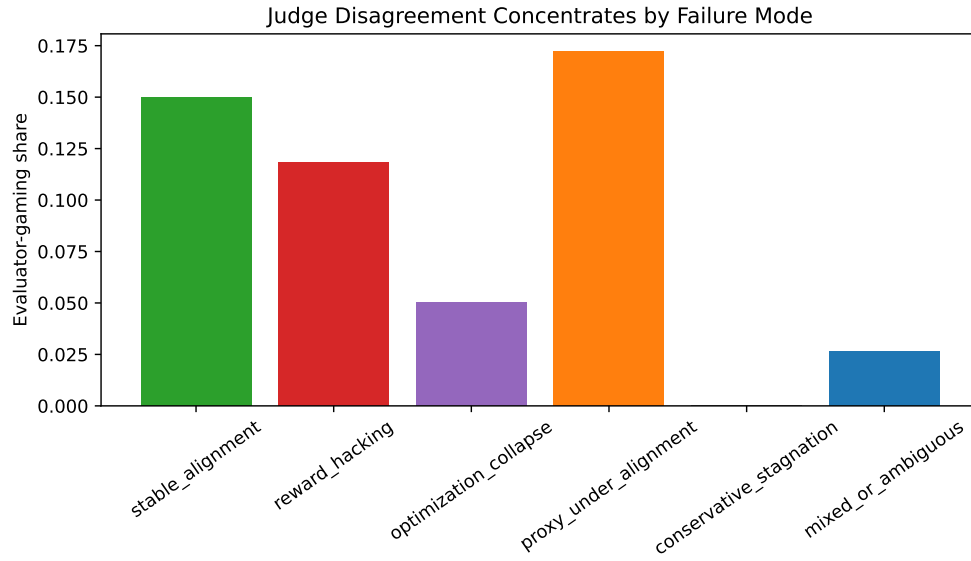


Figure 6: Evaluator-gaming share by row-level failure mode. Judge disagreement is concentrated in specific transition types rather than being uniformly distributed across all rows.